

are being found on how to respond to these users in the best possible way with the best performance response time.

Give us some exciting examples of solutions combining both embedded and cloud computing?

A classic example of embedded and cloud coming together would be from the automotive industry. Traditionally, a car has a lot of ECUs that essentially collect the information from the sensors, provide the necessary information to the driver, and even store the information so that when the car is serviced at a service station, the technicians could retrieve the information for diagnostic purposes. Today, the scenario is that the information must be transmitted in real-time over a 3G or LTE network. The infotainment console is also connected all the time, and the information is sent over to a cloud server. So if there's an issue with the car, a preventive diagnostic could be done.

With this concept of the connected car, we are trying to connect the embedded system within the car to a cloud server in such a way that an analysis could be performed when something is wrong with the vehicle and a real-time feedback could be provided to the driver.

Could you talk about the technical challenges your design and verification team faced?

Testing and verification, in general, is getting more complex today because a wider range of platforms is being supported whereas earlier there were only a few platforms that were supported. This has resulted in the firm having to test each product for multiple platforms.

Apart from the technological challenges, there are also some new challenges on the testing side due to the usage of software applications. For example, earlier the application used to be more PC-based, used either at home or office, but today with the mobile and other devices coming in, the testing moves from an application perspective to the context-related testing.

It's not only about testing the data handling ability of the application but also the unstructured content like video, voice, photo, text messages and others. When a mobile application is being tested, apart from normal functional and non-functional requirements like load, performance and testing, there are other different things that need to be tested, which were absent a few years earlier. We are actively working towards these challenges that are emerging out of both the technology as well as the application.

What changes were made to the development process to add value and improve results?

Currently, there is a lot of work happening in the Agile Model. Agile Development Methodology is something that we have strongly followed in the last four to five years for development and testing to go hand-in-hand all the time. We also try to bring additional value to the customers through test automation—either through the definition of an automation architecture or by bringing on an automa-

tion framework by ourselves, on top of which the test team continues to add the test scripts while the automation team continues to enhance the infrastructure.

Embedded products need specialised testing across multiple layers and at the system level. What difficulties are faced while providing testing services across all these layers?

Embedded products being designed today can be seen as complex special purpose computers that integrate hardware and software from a number of vendors. This includes platform software, network interfaces, security, packet processing, telephony software, graphics and multimedia middleware, JVM, HMI/MMI layer and embedded applications all bundled into a single box.

In terms of architecture, embedded products tend to be very different from each other based on their end use, level of user interactivity and responsiveness desired from the product. Hence, testing such products for various end use scenarios requires sound technical understanding of the complete software stack and domain knowledge. An embedded product test house should be innovative to use or develop the right test jigs and test tools, and exercise great discipline during the test planning and test execution process to cover all product use cases.

What technical challenges do you face while customizing, re-engineering customers' existing products?

In product re-engineering projects, we go back to the drawing board to come up with the best architecture, HW-SW partitioning, power optimisation and user experience for the product. All these factors have to be looked at along with the business challenges such as cost, country-specific regulations and other factors that will make the product successful in the target market.

Can you share your views on new digital TV technologies that are growing along with cloud?

Digital TVs are likely to change from multimedia content receivers to become the central home computer that delivers information and entertainment to all the family members. Technologies that are driving this change are smart connectivity and new cloud services/OTT applications. In the near future, digital TVs will support seamless connectivity to smartphones, tablets and other home devices using technologies such as 802.11n and WiFi Direct. The use of Android in digital TVs, and introduction of new technologies such as Miracast by the WiFi Alliance, are changing the way we interact with TVs. In the future, the digital TV market will be driven by interactivity, i.e., interactive applications that allow consumers to interact with the content, contextual search, contextual advertising and other OTT applications that serve content based on the user profile and interests. ●

(The complete interview can be read at electronicsforu.com)